Regular Expressions Solutions - Intermediate

1. **2010-11**

   Solution: none
   01*001*10*11*001*0 ∪ 01*001*00*111*001*0 ∪ 01*010*111*001*0

2. **2011-12**

   Solution: b, c, d, f

HEX	BINARY	ACCEPTED
a. 48B	010010001011	no
b. 48E	010010001110	yes
c. 5FE	01011111110	yes
d. 67E	011001111110	yes
e. 577	010101110111	no
f. 60E	011000001110	yes

3. **2015-16**

   Solution: As shown
   a (a* ∪ b)* b (b* ∪ a)* = a((a*)*b*)*b((b*)*a*)*
   = a(a*b*)*b(b*a*)*

4. **Mine**

   Solution: B, C, D
   A and E don't work with 3rd b. F doesn't work with 2nd b.

5. **Mine**

   Solution: A, B, C
   D doesn't have an 'l' after the first 'u'. E has no 'u._' and F has no '.' after the I.

6. **Mine**

   Solution: C, E, F
   C and F use the distributive property. F also uses the fact that a*a = aa*.
   E uses the commutative property.

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7. **Mine**

   Solution: A, E, F
   B has 2 digits before the point. C has 3 digits after the point.
   D needs 2 digits after the point.

8. **Mine**

   Solution: D
   A, C, and F don't accept 'abab'. E doesn't accept 'abba'.

9. **Mine**

   Solution: A, C, D, F
   A and F use the first choice. C and D use the second choice.

10. **Mine**

    Solution: A
    The first digit must be a 1 and it must end with 1 or more 0's.